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Application of ChatGPT-based blended medical teaching in clinical education of hepatobiliary surgery

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ABSTRACT

Objective: This study evaluates the effectiveness of incorporating the Chat Generative Pre-trained Transformer (ChatGPT) into the clinical teaching of hepatobiliary surgery for undergraduate medical students.

Materials and methods: A group of 61 medical undergraduates from the Affiliated Hospital of Guizhou Medical University, undergoing hepatobiliary surgery training, were randomly assigned to either an experimental group (31 students) using ChatGPT-based blended teaching or a control group (30 students) with traditional teaching methods. The evaluation metrics included final exam scores, teaching satisfaction, and teaching effectiveness ratings, analyzed using SPSS 26.0 (SPSS Inc., Chicago, IL) with *t*-tests and χ^2 tests.

Results: The experimental group significantly outperformed the control group in final exam theoretical scores (86.44 ± 5.59 vs. 77.86 ± 4.16 , $p < .001$) and clinical skills scores (83.84 ± 6.13 vs. 79.12 ± 4.27 , $p = .001$). Additionally, the experimental group reported higher teaching satisfaction (17.23 ± 1.33) and self-evaluation of teaching effectiveness (9.14 ± 0.54) compared to the control group (15.38 ± 1.5 and 8.46 ± 0.70 , respectively, $p < .001$).

Conclusions: The integration of ChatGPT into hepatobiliary surgery education significantly enhances theoretical knowledge, clinical skills, and overall satisfaction among medical undergraduates, suggesting a beneficial impact on their educational development.

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KEYWORDS

Chat Generative Pre-trained Transformer; clinical teaching in hepatobiliary surgery; medical students

1. Introduction

Medical education plays a pivotal role in cultivating medical professionals. Traditional teaching methods, however, are becoming insufficient in meeting the evolving demands of medical education (Rajkomar et al. 2019; Thirunavukarasu et al. 2023). The rapid development of information technology, with the widespread application of artificial intelligence (AI) and natural language processing (NLP), is infiltrating various fields, inevitably bringing new opportunities and challenges to medical education (Esteve et al. 2019; Jowsey et al. 2023). Chat Generative Pre-trained Transformer (ChatGPT), as a large language model, signifies the forefront of open AI development. Its impact on the field of clinical medicine and its potential role in medical education are garnering increasing attention (Varghese and Chapiro 2023). Studies have shown that AI models can provide substantial support in clinical practice, encompassing data analysis, clinical research, and assistance in clinical guideline decision-making (Obermeyer and Emanuel 2016; Jiang et al. 2017). They also extend to areas like clinical trial recruitment, clinical data management, research support, and patient education (Shaban-Nejad et al. 2018).

Blended medical teaching reforms that integrate the powerful language model ChatGPT with traditional medical education aim to provide students with personalized,

Practice points

- Integrating Chat Generative Pre-trained Transformer (ChatGPT) into the clinical teaching of hepatobiliary surgery significantly improves students' theoretical knowledge and clinical skills.
- The use of ChatGPT-based blended teaching methods results in higher levels of teaching satisfaction among students.
- The positive self-evaluation scores of teaching effectiveness from students highlight the effectiveness of ChatGPT-based teaching methods.
- The study results emphasize the statistical robustness and educational impact of incorporating ChatGPT into medical education.
- This study highlights the potential of blended teaching models in the clinical education of hepatobiliary surgery, providing insights for teaching in other medical fields.

interactive learning experiences through the aid of machine intelligence, thereby driving innovation and enhancement in medical education (Topol 2019). In light of this context, this study aims to apply ChatGPT in medical education to deliver personalized teaching content and learning guidance, along with real-time learning feedback, to foster deeper learning and reflection among students. This initiative is anticipated to cultivate medical professionals with innovative capabilities and problem-solving skills.

2. Methods

2.1. Study subjects

This was a prospective, randomized, controlled study. This study selected 61 clinical undergraduate medical students who undertook their internship in hepatobiliary surgery at the Affiliated Hospital of Guizhou Medical University in the fall semester of 2023. A simple randomization was adopted for this study. All students were renumbered as 1 to N . If the assigned number was odd, he/she entered the experimental group, whereas if the number was even, he/she entered the control group. The experimental group (31 students) was taught using a ChatGPT-based blended teaching approach (OpenAI, San Francisco, CA), while the control group (30 students) received traditional teaching. Each group was supervised by teaching staff consisting of one instructor and three assistants who held full-time professional positions within the Department of Hepatobiliary Surgery. An overview of the flow chart of the study design is presented in Figure 1.

2.2. Teaching methods

2.2.1. Control group teaching

According to the requirements of the hepatobiliary surgery internship teaching syllabus, clinical teachers in the department meticulously prepared relevant materials (selecting typical cases from the department and creating courseware that meets the syllabus requirements and closely relates to actual cases). In class, teachers first delivered theoretical knowledge or conducted operational demonstrations,

followed by students asking questions and, in groups, organizing related knowledge around cases, consulting textbooks, reference books, and literature. The ward internship included daily rounds, patient history inquiries, and other clinically relevant tasks.

2.2.2. Experimental group teaching

In addition to the control group's teaching content, the experimental group primarily incorporated the ChatGPT language model, using a ChatGPT-based deep integration model to replace traditional 'typical case' teaching and post-class discussions. This method also integrated medical ethics and humanities education. During the internship, instructors conducted real-time interactive teaching for common diseases and routine operations, guiding interns to use ChatGPT for independent learning, posing questions, engaging in simulated dialogues and discussions, and incorporating literature reviews suggested by ChatGPT to analyze and summarize diseases for preliminary diagnoses, differential diagnoses, and treatment plans. Additionally, ChatGPT assisted in clarifying the focus of physical examinations, key points in history taking, and important considerations in clinical operations, and analyzing the differences between these and clinical reality, integrating medical humanities knowledge and involving patients in the teaching process to enhance doctor-patient communication efficiency and satisfaction.

2.3. Evaluation method

After four weeks of learning and practice, a panel consisting of three or more attending physicians evaluated the

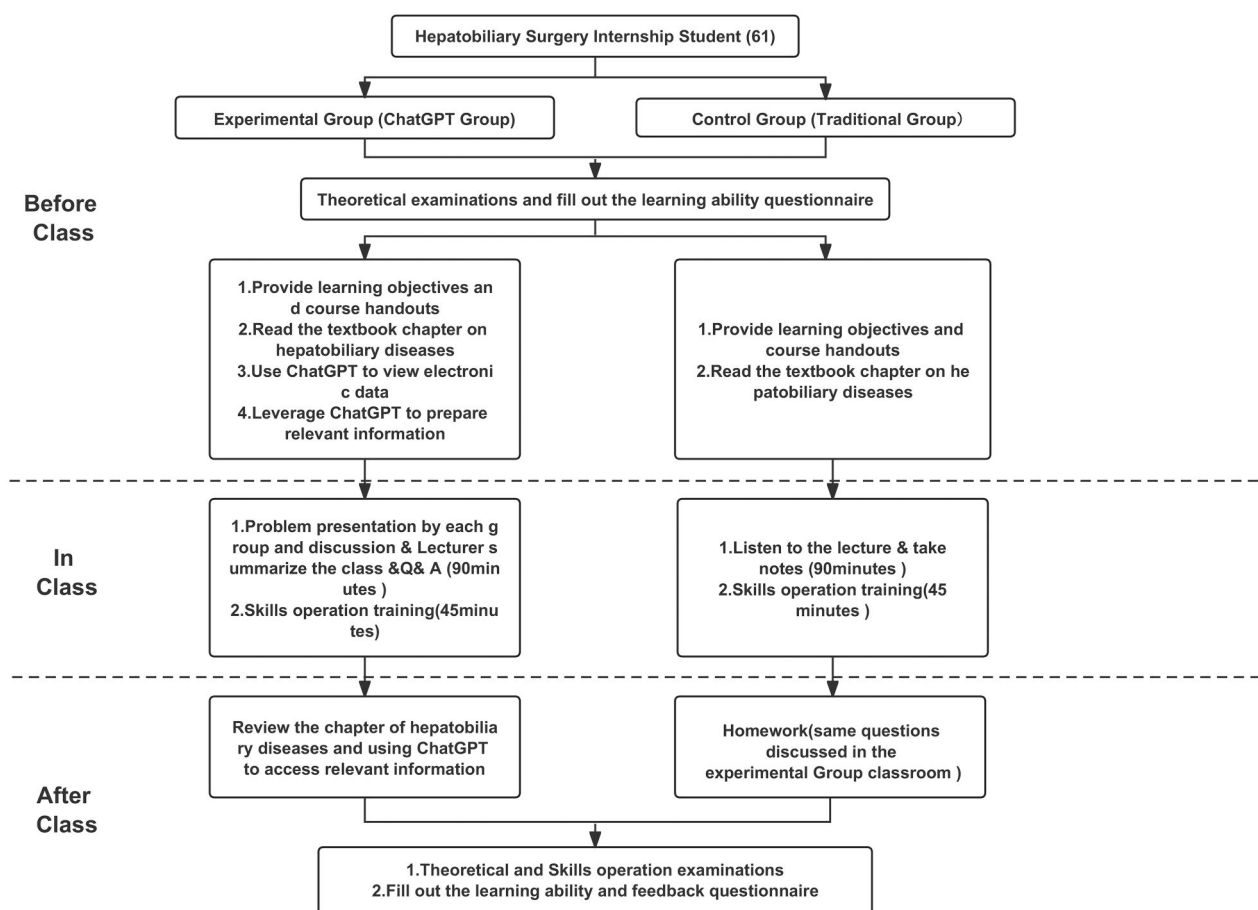


Figure 1. An overview of the study design.

interns. All objective evaluations totaled 100 points to assess the student's basic knowledge of the disease. Additionally, the students were asked to report the time required for pre-class preparation and post-class review.

2.3.1. Objective evaluation

Clinical skills of interns were assessed based on their performance in the process and content of history taking, physical examination, initial diagnosis, differential diagnosis, treatment plan, and communication with patients.

A standardized hepatobiliary surgery theory examination was used to objectively evaluate the interns' grasp of relevant conceptual knowledge. At three days before class, all students conducted a pre-class theoretical examination, followed by a post-test after three days of the class.

2.3.2. Subjective evaluation

A questionnaire survey was conducted to evaluate the interns' satisfaction with teaching and self-assessment of the teaching effectiveness. Students were asked to select the answer that best described their actual situation and were told that the questionnaire had no relationship with their academic performance. Furthermore, all students completed a feedback questionnaire three days after class (Additional file 1).

2.4. Evaluation scales

The teaching satisfaction survey was a self-designed questionnaire, including 10 items: improvement in knowledge mastery, enhancement in diagnostic ability, expansion of clinical thinking, improvement in doctor-patient communication, enhancement in problem-solving ability, increase in learning interest and initiative, improvement in team communication and expression, enhancement in team cooperation, achievement of teaching goals, and willingness to continue using this teaching method. Scoring criteria: yes = 2 points, unsure = 1 point, no = 0 points.

The self-assessment of teaching effectiveness survey was also a self-designed questionnaire, including five items: improvement in communication skills, self-learning ability, language expression, summarization and induction ability, and logical thinking. Scoring criteria: yes = 2 points, unsure = 1 point, no = 0 points. Scores were allowed to two decimal places.

2.5. Statistical methods

Statistical analysis was conducted using SPSS 26.0 software (IBM Corp., Armonk, NY). Quantitative data were expressed as mean $\pm$ standard deviation, and group comparisons were made using the *t*-test; categorical data were expressed as percentages and analyzed using the χ^2 test. The level of significance was set at $\alpha = 0.05$.

3. Results

3.1. Basic characteristics and information

The class attendance and questionnaire response rates of both groups were 100%. There was no statistical significance in the general information such as gender, age, and subject baseline scores between the two groups of interns ($p > .05$), indicating comparability. The subject baseline scores are shown in Table 1.

3.2. Comparison of final assessment scores and clinical skills between the two groups of interns

According to the statistical results, the final theoretical assessment scores of the interns in the experimental group were (86.44 ± 5.59), which was significantly higher than that of the control group (77.86 ± 4.16), with a statistically significant difference ($p < .001$). The clinical skills assessment scores of the experimental group were (83.84 ± 6.13), also surpassing those of the control group (79.12 ± 4.27), with a significant difference ($p = .001$), as shown in Table 2.

3.3. Comparison of teaching satisfaction and self-assessment of teaching effectiveness between the two groups of interns

According to the statistical results, the teaching satisfaction survey scores of the interns in the experimental group were (17.23 ± 1.33), which were significantly higher compared to the control group (15.38 ± 1.54) with a significant difference ($p < .001$). The self-assessment scores of teaching effectiveness for the interns in the experimental group were (9.14 ± 0.54), again significantly higher than those in the control group (8.46 ± 0.70), with a significant difference ($p < .001$), as shown in Table 3.

Table 1. Basic characteristics of two groups.

Group	Gender		Age (years)	Subject baseline (scores)
	Male	Female		
Control group	16	14	22.08 ± 0.93	60.75 ± 5.01
Experimental group	17	14	22.16 ± 0.87	59.88 ± 4.66
<i>t</i> / χ^2 -value	0.014		0.312	0.275
<i>p</i> Value	.906		.760	.485

Table 2. Comparison of theoretical assessment and clinical skills assessment scores.

Group	Theoretical score	Clinical skills score		
		PE	PO	Total score
Control group	77.86 ± 4.16	39.49 ± 2.90	39.63 ± 2.70	79.12 ± 4.27
Experimental group	86.44 ± 5.59	42.18 ± 3.54	41.65 ± 3.49	83.84 ± 6.13
<i>t</i> -value	6.785	3.246	2.522	3.473
<i>p</i> Value	.000	.002	.014	.001

PE: physical examination; PO: practical operation.

Table 3. Comparison of teaching satisfaction and self-assessment of teaching effectiveness.

Group	Teaching satisfaction survey	Self-assessment of teaching effectiveness
Control group	15.38 ± 1.54	8.46 ± 0.70
Experimental group	17.23 ± 1.33	9.14 ± 0.54
<i>t</i> -value	5.042	4.267
<i>p</i> Value	.000	.000

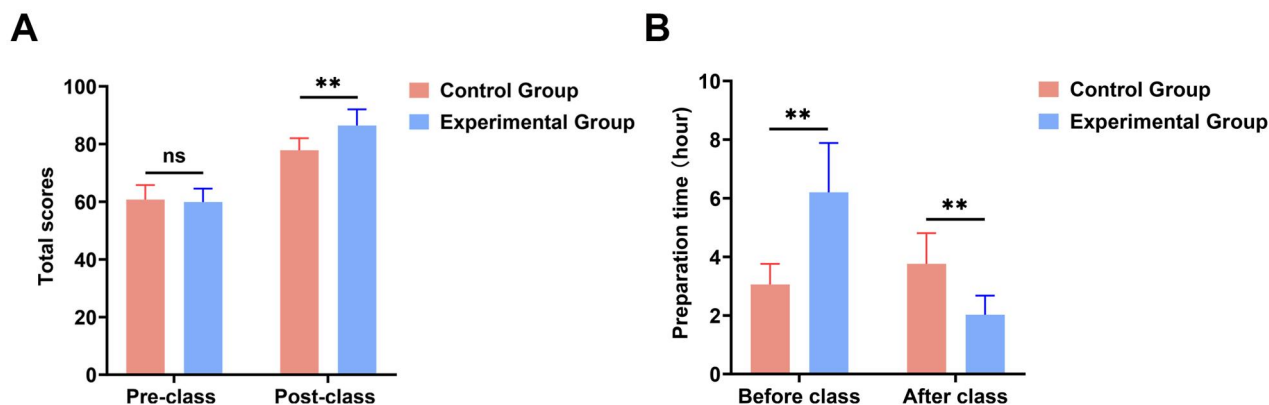


Figure 2. Comparison of pre- and post-course scores and study time between two groups.

3.4. Comparison of pre- and post-course study time and pre- and post-teaching grades between two groups of interns

It is to be noted that there was no significant difference in the pre-class theory scores between the experimental and control groups ($p > .05$). However, after the course, the post-class theory scores of the experimental group were significantly higher than those of the control group ($p < .05$) (Figure 2(A)). In addition, we compared the preparation time before and after the class between the experimental and control groups. Overall, the pre-course preparation time of the experimental group was significantly longer than that of the control group ($p < .05$), while the post-course preparation time of the experimental group was significantly shorter than that of the control group ($p < .05$) (Figure 2(B)).

4. Discussion

The importance of medical education in nurturing future medical professionals cannot be overstated (Lee 2023). However, with the continuous development in the field of medicine, traditional teaching methods have gradually become inadequate. Fortunately, with the rapid advancement of information technology, the wide application of AI and NLP technologies has begun to unfold new opportunities (Alqahtani et al. 2023). ChatGPT, supported by the large language model GPT-3.5, is leading the reform in medical education, bringing novel possibilities to the field of clinical medicine (Heng et al. 2023).

Research has clearly indicated that the application of AI models in the field of medicine extends beyond data analysis and clinical research, to include clinical decision support, clinical trial recruitment, clinical data management, research support, and patient education (Shen et al. 2023). These applications open up more possibilities and opportunities for medical education (Gilson et al. 2023; Scherr et al. 2023). The core idea of blended medical teaching reform is to integrate the advanced language model ChatGPT with traditional medical teaching to provide a personalized, interactive learning experience, thus driving innovation and enhancement in medical education (Morales-Ramirez et al. 2023). The goal of this reform is not only to cultivate medical professionals proficient in their specialty but also to foster doctors with a medical humanistic spirit, who care for patients, respect life, and possess innovative and problem-solving abilities (Han et al. 2023; Hwang 2023; Sahu et al. 2023).

In this study, we selected 61 clinical undergraduate medical students from the Affiliated Hospital of Guizhou Medical University as subjects, dividing them into an experimental group and a control group, and compared a ChatGPT-based blended medical teaching method with traditional teaching methods. Although the preparation time of the experimental group was longer before class, its review time was also significantly shorter after class. The results showed that the experimental group achieved significant advantages in hepatobiliary surgery internships, including theoretical assessment scores and clinical skills assessment scores. These results suggest that the blended medical teaching method has clear advantages in improving students' grasp of theoretical knowledge and clinical skills. Additionally, the results from the teaching satisfaction surveys and self-assessment of teaching effectiveness indicate that students in the experimental group showed higher satisfaction and self-evaluation scores. This reflects their positive attitude and recognition towards the ChatGPT-based blended medical teaching method.

The findings of this study underscore the potential of the ChatGPT-based blended medical teaching approach, particularly in clinical teaching of hepatobiliary surgery. This method not only provides a higher level of personalized learning and interactive experience but also helps cultivate medical professionals with stronger clinical skills and medical humanistic values. However, this study has limitations, such as a relatively small sample size and being limited to a specific medical field, necessitating further research to validate this method's applicability across different medical disciplines and educational institutions of varying sizes.

In conclusion, the ChatGPT-based blended medical teaching approach brings new opportunities and challenges to medical education. It not only improves students' academic performance but also cultivates their clinical skills and medical humanistic literacy. As AI technology continues to evolve, we can anticipate this approach playing a more significant role in future medical education, contributing to the training of more excellent medical professionals.

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Author contributions

Changhao Wu: writing of the manuscript; Liwen Chen, Min Han, and Nenghong Yang: data collection and organization; Zhu Li: data analysis and organization; Chao Yu: revision of the manuscript and providing suggestions.

Ethics statement

This study was approved by the Ethical Review at the Department of Hepatic-Biliary Pancreatic Surgery, The Affiliated Hospital of Guizhou Medical University (Guiyang, China).

Consent form

Written informed consent was obtained from all participating students.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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